

Door Entry & Access Control Standard

3D Face Recognition, SIP Intercom, and Secure Strike Integration

Date: January 23, 2026
Version: v1.4
Status: Prescriptive Specification

1. System Overview

The project utilizes an IP-based video intercom and access control system centered on **Akuvox** 3D face recognition hardware. All control and power logic is centralized in a dedicated **DIN-rail LCP (LCP-MDF)** located in the Machine Room.

2. Hardware Specification

2.1 Outdoor & Indoor Units

Location	Model	Features	Power
Front Door	Akuvox X915S	8" Touch, 3D Face Rec, IK10	PoE + 12V DC (Backup)
Service Doors	Akuvox E16C	5" Screen, 3D Face Rec, Dual Cam	PoE + 12V DC (Backup)
Hallway Tablet	Akuvox S567G	10" Android 12, Wi-Fi 6	PoE

- Tablet Placement:** The **S567G** monitor must be wall-mounted in the **Office Hallway area** at 58" AFF.

2.2 Secure Logic & Locking (Centralized)

All following components live in **LCP-MDF**:

- Secure Relays:** **Akuvox SR01** (One per door). Mounted to DIN rail via adapter clips.
- KNX Binary Input:** For power monitoring (AC Fail / Low Bat).
- Electric Strike:** **HES 1006** (Fail-Secure). Configured for **24V DC**.
- Power Controller:** **Altronix eFlow6N**. Mounted to DIN rail via Altronix DIN-mount kit.
- Battery Backup:** **Two (2) 12Ah SLA Batteries** (Series wired for 24V). Placed at the base of the LCP-MDF enclosure.

3. Machine Room LCP-MDF Configuration

3.1 Mounting Standard

The **LCP-MDF** must be a professional DIN-rail enclosure (e.g., Saginaw or similar) to match the project's electrical standards.

- Rail 1 (Top):** Power distribution (Altronix board + Fuses).
- Rail 2 (Middle):** Secure Relays (SR01s) and KNX Binary Input.
- Base (Bottom):** Battery storage.

3.2 Power Outage Notification

The **Altronix AC-Fail** and **Low-Bat** dry contacts are wired into the **KNX Binary Input**.

- **Logic:** If utility power is lost, Home Assistant triggers an immediate "Power Outage" alert to the Owner's phone.
-

4. Connection Map (The "Nervous System")

4.1 Home-Runs to LCP-MDF

1. **Cat6/6a:** From each Intercom/Tablet to the **UniFi PoE Switch** in the rack.
 2. **18/4 Shielded:** From each Intercom to its assigned **SR01** in LCP-MDF.
 - Pair 1: RS485 Data (Encrypted).
 - Pair 2: 12V DC Backup Power.
 3. **18/2 (or 14/2):** From each **SR01** in LCP-MDF to its assigned **HES Strike**. (24V Lock Power).
-

5. Security & Installation

- **Pairing:** Intercoms must be software-paired to SR01s to ensure the encrypted link is active.
 - **Firewall:** Garage flush-mounts require **STI SpecSeal Putty Pads**.
 - **Waterproofing:** Silicone seal top and sides; leave bottom weep-hole open.
-

End of Specification.